LENS: Likelihood Evaluation and Neighbor Smoothing for Attribute-Based Person Retrieval on Unseen Cameras

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Abstract

Attribute-based person retrieval ranks surveillance images from a binary attribute query, yet mean Average Attribute Degree of Match (mADM) drops on unseen cameras as predicted probabilities shift. Standard pipelines rank by L1 distance, which penalizes a confident single-attribute mismatch no more than small uncertainties across many attributes, while source-fitted Platt scaling lowers average held-out negative log-likelihood yet hurts retrieval mADM as per-attribute logit biases shift across cameras. We present LENS (Likelihood Evaluation and Neighbor Smoothing), an empirical study of how retrieval scoring, training-time group consistency, and test-time gallery smoothing interact under camera shift. Combining fine-tuned DINOv3 and CLIP teachers with a distilled student, LENS regularizes mutually exclusive attribute groups with group cross-entropy, scores gallery images by the exact-query log-likelihood, and smooths logits over nearest neighbors using DINOv3 patch features from the same pass. Across the high-resource held-out folds (PETA and Market-1501), these choices raise mADM for single models and ensembles. Likelihood scoring beats L1 across all 48 development model configurations, kNN smoothing improves all 46 tested configurations, and LENS reaches 68.17 mADM on the UPAR 2027 hidden test set.

1. Introduction

Attribute-based person retrieval finds people in surveillance cameras from a binary query of gender, age, clothing, and accessories without a query image [27]. Most pipelines predict attribute probabilities per gallery image and compare them with the query [25, 27]. On unseen cameras, changes in viewpoint, lighting, resolution, and annotation rules shift predicted probabilities [27]. In the 2023 and 2024 UPAR challenges [6, 7] across PA100k [18], PETA [8], and Market-1501 [16, 35], several teams beat the attribute recognition baseline [4], yet none beat the retrieval baseline [6, 7].

Existing UPAR pipelines rank gallery images by L1 distance between the binary query and predicted probabilities [31], and the 2024 report suggested that classifiers become miscalibrated on new cameras [7]. We test both hypotheses under a leave-one-dataset-out protocol. First, L1 distance sums errors linearly, so a confident mismatch on one queried attribute adds no more penalty than small uncertainties across several attributes. Ranking by the exact-query log-likelihood [14, 23] penalizes near-zero probabilities on queried attributes and raises mADM on the exact same probabilities without retraining. Second, decomposing NLL into calibration and refinement [12] provides an in-sample diagnostic showing substantial held-out miscalibration, yet source per-attribute logit biases are nearly uncorrelated with held-out biases, potentially associated with prevalence, rare positives, and annotation conventions. Consequently, source-fitted Platt scaling lowers average held-out NLL on PETA (6.20 to 5.94 nats) while lowering mADM (69.53 to 68.51), showing that improving an average proper scoring rule does not guarantee better query-conditioned ranking under bias shift.

Because probability shifts cannot be fixed post-hoc from source validation data in our setting, we present LENS (**L**ikelihood **E**valuation and **N**eighbor **S**moothing, Fig. 1) as an empirical study of how retrieval scoring, training-time group consistency, and test-time gallery structure interact on unseen cameras. Evaluated on the high-resource held-out folds (PETA and Market-1501), LENS pairs fine-tuned DINOv3 [24] and CLIP [21] teachers with Pre-LN cross-attention query heads and a distilled student [2, 11], while CLIP adds a text score and WiSE-FT [32]. We regularize mutually exclusive groups with group cross-entropy in training, and apply transductive kNN smoothing over gallery neighbors at inference using DINOv3 patch features from the same pass [30]. Our main findings are

- Exact-query log-likelihood scoring beats L1 distance on all six metrics for our ensemble and across all 48 tested model configurations.
- Parameter swaps show that while Platt scale transfers

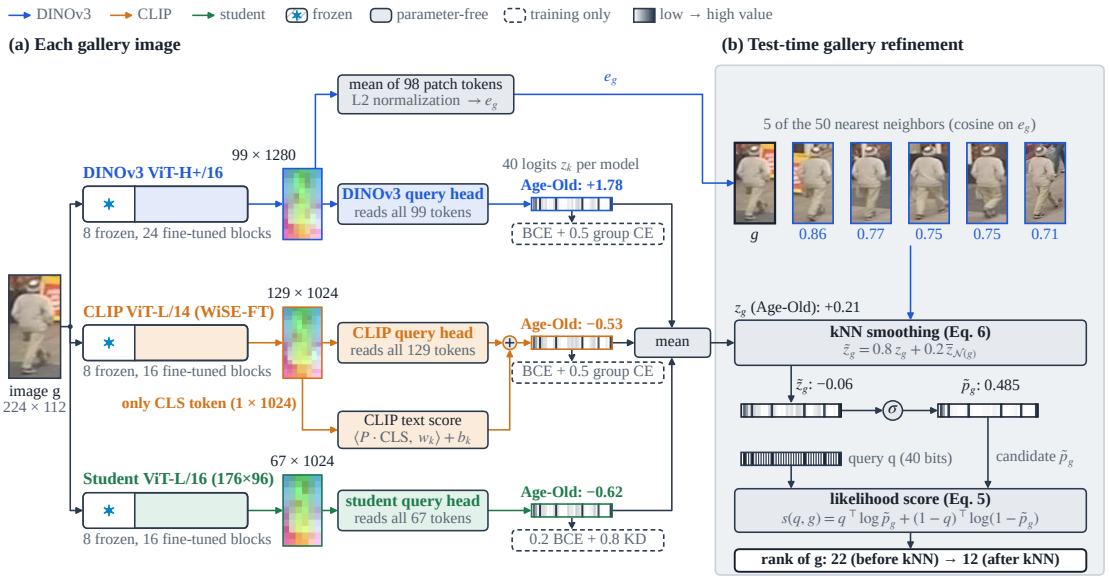


Figure 1. Overview of the LENS retrieval pipeline. (a) Three attribute models process one gallery image g from the held-out PETA fold. Each encoder freezes the patch embedding and first eight blocks (snowflake icon) and fine-tunes the remaining blocks with a query head (Fig. 2). The DINOv3 and CLIP models use BCE and group CE losses, and the student is distilled from two teachers. At inference, their logits are averaged into z_g, while the L2-normalized mean of the 98 patch tokens of the DINOv3 model forms e_g. (b) Five of the $M = 50$ nearest neighbors of g by cosine similarity of e_g (here showing the same person in other frames) are averaged in kNN smoothing (Eq. (6)), flipping the Age-Old logit sign to match the label and improving the rank of g under Eq. (5) from 22 to 12.

across domains, per-attribute biases shift so source maps hurt mADM even as average NLL drops.

- Regularizing mutually exclusive groups with group cross-entropy reduces within-group conflicts and raises mADM across single models and ensembles.
- Transductive kNN smoothing over DINOv3 neighbors improves all 46 tested model configurations without training, with gains governed by gallery redundancy.

2. Related work

Attribute-based person retrieval. Person retrieval by attributes matches surveillance images against a binary query without a query image [25, 27]. Most pipelines predict gallery probabilities and compare them with the query [26, 27]. Specker and Beyerer introduced recalibrated distance weighting and mADM [25, 26], while the UPAR baseline ranks by L1 distance [7, 31], which sums errors linearly and penalizes a confident mismatch no more than small uncertainties across many attributes. Under conditional independence and calibrated probabilities, the probability ranking principle [23] orders candidates by the probability of matching the query, though mADM also credits near matches and pedestrian attributes include mutually exclusive groups. Direct Attribute Prediction (DAP) [14] divides by class priors, whereas LENS uses the exact-query log-likelihood without prior division, beating L1 and matching expected near-match corrections on PETA.

Calibration and attribute dependencies. Because log-likelihood heavily penalizes near-zero probabilities on queried attributes, it relies on probability quality under shift. The 2024 UPAR report noted that models become miscalibrated on unseen cameras [7]. While post-hoc Platt [20] or temperature scaling [10] adjusts logits on validation data, calibration maps degrade under shift [19]. Decomposing NLL into calibration and refinement [12] shows that while held-out cameras suffer miscalibration, per-attribute biases shift across datasets, potentially associated with prevalence and annotation differences, so source calibration hurts retrieval. Rather than adjusting logits post-hoc, LENS enforces consistency in training. Whereas BCE [15] treats all 40 attributes independently, we regularize mutually exclusive groups (age, hair length, clothing colors [27]) with group cross-entropy.

Test-time gallery smoothing. Single-frame predictions remain sensitive to viewpoint, blur, or occlusion. In image retrieval and re-identification, transductive methods refine rankings using gallery structure, including k-reciprocal re-ranking over mutual neighbors [36], query expansion over top matches [5], and database-side neighbor averaging [1, 9, 29]. Yet these require query and gallery images to share a visual feature space, absent when the query is binary text. For classification under shift, LAME [3] adapts predictions via graph optimization, whereas LENS reuses DINOv3 tokens from the same pass for kNN smoothing.

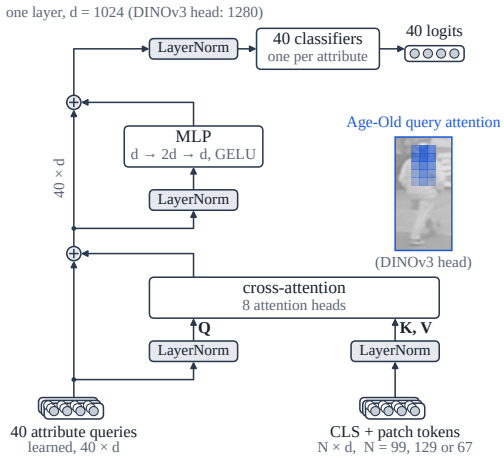


Figure 2. Pre-LN cross-attention query head with width d (1280 for DINOv3, 1024 for CLIP and student, Sec. 4). The inset shows Age-Old attention weights over the 14×7 grid of g in Fig. 1.

3. Method

Task and evaluation metric. Each query is a binary vector $q \in \{0,1\}^K$ over K attributes. Under the UPAR protocol [26, 27], each distinct ground-truth label vector in the gallery forms a query, and an image $g \in \mathcal{G}$ with binary labels $y_g \in \{0,1\}^K$ is an exact match when $y_g = q$. Let $\mathrm{dom}(q,g) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{I}(y_{gk} = q_k)$ be the share of attributes where g agrees with q, normalized relative to the gallery mean $\overline{\mathrm{dom}}_q = \frac{1}{|\mathcal{G}|}\sum_{g' \in \mathcal{G}} \mathrm{dom}(q,g')$ as

$$\mathrm{ndom}(q,g) = \max\left(0, \frac{\mathrm{dom}(q,g) - \overline{\mathrm{dom}}_q}{\max_{g' \in \mathcal{G}} \mathrm{dom}(q,g') - \overline{\mathrm{dom}}_q}\right). \quad (1)$$

Let $r_1 < r_2 < \cdots < r_R$ be the ranks of the R exact matches in $\mathcal{G}$, and $g_{(j)}$ the image at rank j. Average Attribute Degree of Match (ADM) is defined as

$$\mathrm{ADM}(q) = \frac{1}{R}\sum_{m=1}^{R} \frac{1}{r_m}\sum_{j=1}^{r_m} \mathrm{ndom}\big(q, g_{(j)}\big), \quad (2)$$

and mADM is the mean of $\mathrm{ADM}(q)$ over queries, alongside mAP, mINP, Rank-1, Rank-5, and Rank-10 (in points). Unlike mAP, which gives zero credit to near matches, mADM evaluates at exact-match ranks r_m while crediting near matches ranked ahead via $\mathrm{ndom}(q, g_{(j)})$. Thus, a confident error on an exact match or an unpenalized distractor mismatch pushes exact matches down. LENS addresses this (Fig. 1) by learning non-conflicting logits z_g, ranking by exact-query log-likelihood rather than L1, and smoothing single-frame logit noise over gallery neighbors.

Attribute recognition. Predicting reliable attribute logits on unseen cameras faces two demands. Small region attributes (footwear, headwear, bags) need fine localization, whereas clothing colors and global attributes need semantic invariance across cameras. Global average pooling blurs small regions [15], and no single encoder meets both needs. Self-supervised DINOv3 [24] learns localized patch tokens that separate fine details, whereas CLIP [21] learns from web captions for stronger semantic and color invariance. LENS pairs fine-tuned DINOv3 and CLIP models (Fig. 1a). Each freezes its patch embedding and first eight blocks (snowflake icon) and fine-tunes remaining blocks with a one-layer Pre-LN cross-attention query head [17, 22] (Fig. 2), where K learned queries attend to image regions (Fig. 2 inset) before an MLP and linear classifiers output visual logits $z^{\mathrm{vis}} \in \mathbb{R}^K$. To retain language alignment without overfitting the text encoder, CLIP adds a text score to its visual logit (Fig. 1a) as

$$z_{\mathrm{C},k} = z^{\mathrm{vis}}_{\mathrm{C},k} + \langle P \cdot \mathrm{CLS},\, w_k \rangle + b_k, \quad (3)$$

where P is the frozen projection, $w_k = t_k - t_{\mathrm{neutral}}$ is the prompt-ensembled attribute embedding minus "a person", and b_k is a learnable bias initialized from log-prior odds. Because fine-tuning distorts pretrained features [13], WiSE-FT [32] interpolates CLIP weights θ_{ft} with θ_0 via $\theta = (1-\lambda)\theta_0 + \lambda\theta_{\mathrm{ft}}$ ($\lambda \in [0,1]$).

While pairing DINOv3 and CLIP strengthens features, class-balanced BCE [15], $\mathcal{L}_{\mathrm{bce}} = -\sum_{k=1}^{K} \omega_k [y_k \log \sigma(z_k) + (1-y_k)\log(1-\sigma(z_k))]$, treats the K attributes as independent tasks on hard labels. Independent sigmoids let mutually exclusive attributes (competing ages or clothing colors) fire at once, and hard labels on occluded crops cause overconfident errors. To curb conflict within the S mutually exclusive groups $\mathcal{A} = \{A_1, \ldots, A_S\}$ ($\sum_{k \in A_s} y_k \le 1$), we regularize both models with categorical group cross-entropy over each group $A \in \mathcal{A}$ with a unique active attribute $y_A \in A$ via

$$\mathcal{L}_{\mathrm{gce}} = -\sum_{A \in \mathcal{A}} \mathbb{I}\big(\textstyle\sum_{k \in A} y_k = 1\big) \log \frac{\exp(z_{y_A})}{\sum_{j \in A} \exp(z_j)}, \quad (4)$$

where $\mathbb{I}(\sum_{k \in A} y_k = 1)$ skips groups without a unique positive (unlabeled or multi-color clothing), optimizing $\mathcal{L} = \mathcal{L}_{\mathrm{bce}} + \gamma \mathcal{L}_{\mathrm{gce}}$ ($\gamma > 0$). To soften overconfident errors and add a third model, we distil a DINOv3 student [11] from DINOv3 and CLIP teachers on mixup [34] and CutMix [33] images [2], minimizing $\mathcal{L}_{\mathrm{student}} = (1-\beta)\mathcal{L}_{\mathrm{label}} + \beta \mathcal{L}_{\mathrm{KD}}$ against $\bar{p}_t = \sigma(\frac{1}{2}(z_{\mathrm{D}} + z_{\mathrm{C}}))$. At inference, the three logits are averaged into $z_g = \frac{1}{3}(z_{\mathrm{D}} + z_{\mathrm{C}} + z_{\mathrm{S}}) \in \mathbb{R}^K$ (Fig. 1a), with $p_g = \sigma(z_g)$.

Exact-query likelihood scoring. Given $p_g = \sigma(z_g)$ for each $g \in \mathcal{G}$, standard pipelines rank by L1 distance $d(q,g) = \sum_{k=1}^{K} |q_k - p_{gk}|$ [31]. Because $d(q,g)$ sums errors linearly, it cannot separate one confident mismatch from many small uncertainties. A distractor matching $K-1$ attributes that is confidently wrong on one queried attribute ($p_{gk} \approx 0$ for $q_k = 1$) adds less than 1 to $d(q,g)$ and ranks ahead of an exact match with slight uncertainty across

Fold	Score	mADM	mAP	mINP	R-1	R-5	R-10
PETA	L1 distance [31]	62.88	24.03	18.67	21.75	39.59	47.22
	group-softmax	68.62	35.47	28.77	**32.10**	54.34	**63.41**
	likelihood (5)	**69.91**	**35.51**	**29.08**	31.57	**54.68**	63.07
Market	L1 distance [31]	69.29	25.08	8.97	32.53	51.46	58.00
	group-softmax	73.63	34.38	16.57	**39.64**	**61.92**	**71.97**
	likelihood (5)	**74.90**	**34.39**	**16.84**	37.97	61.30	70.87

Table 1. Scoring rules on DINOv3 + CLIP (CLIP with text score and WiSE-FT), in points (group-softmax uses the single-seed logits of Tab. 4, where Eq. (5) reaches 69.84 and 74.59).

several attributes. Because mADM (Eq. (2)) evaluates at exact-match positions r_m, a probabilistic alternative orders candidates by the probability that all K attributes match q at once [14, 23]. Direct Attribute Prediction (DAP) [14] divides posteriors by class priors π_k. In log-space, the prior term depends only on q and cancels across $g \in \mathcal{G}$, whereas shifting logits by source log-priors before $\sigma(\cdot)$ distorts $\sum_{k=1}^{K} \log(1 - p_{gk})$ under prior shift. LENS ranks by the exact-query log-likelihood against the 40-bit query q (Fig. 1b) as

$$s(q, g) = \sum_{k=1}^{K} \left[q_k \log p_{gk} + (1 - q_k) \log(1 - p_{gk}) \right], \quad (5)$$

which simplifies via $z_{gk} = \log \frac{p_{gk}}{1-p_{gk}}$ to $s(q, g) = q^\top z_g + \sum_{k=1}^{K} \log(1 - p_{gk})$, one matrix product $QZ^\top$ plus one constant per image. Two caveats apply to Eq. (5). First, mADM credits near matches via $\mathrm{ndom}(q, g_{(j)})$, so exact-match ranking is not guaranteed to be mADM-optimal. Second, summing marginal log-probabilities assumes conditional independence whereas $\mathcal{L}_{\mathrm{gce}}$ (Eq. (4)) regularizes mutually exclusive logits without forming a normalized joint distribution. On PETA with ViT-B, adding exact expected ndom adds only $+0.29$ points (Sec. 4), and keeping independent sigmoids in Eq. (5) penalizes near-zero probability on any queried attribute without forcing a single positive.

kNN smoothing. Because Eq. (5) heavily penalizes confident errors, an exact match g whose logit on one attribute is flipped by single-frame occlusion, blur, or angle suffers a steep rank drop. Transductive gallery refinement (Fig. 1b) addresses this under the batch setting permitted by the UPAR Track 2 automatic re-ranking rules [30], where a gallery image g shares visual neighbors from other frames of the same person and different people in matching clothing. Averaging logits across visually consistent neighbors suppresses single-frame noise without labels or queries. Rather than running a separate network, we reuse the patch tokens of the DINOv3 encoder from the same pass (Fig. 1a), defining e_g as the L2-normalized mean of its 98 patch tokens. Let $\mathcal{N}(g)$ be the M nearest neighbors of g in $\mathcal{G} \setminus \{g\}$ by cosine similarity of e_g (Fig. 1b). We smooth

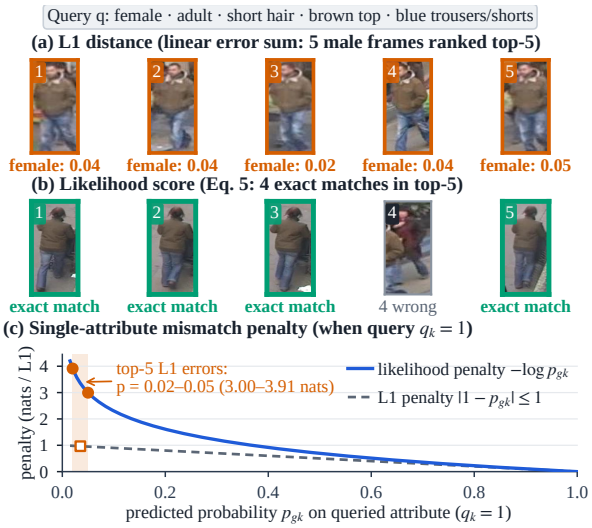


Figure 3. L1 distance vs. likelihood score on one PETA query using the DINOv3 + CLIP ensemble of the PETA fold. (a) L1 distance ranks five male frames first ($p_{\mathrm{female}} \in [0.02, 0.05]$). (b) The likelihood score places four exact matches in the top five (ranks 1, 2, 3, and 5, where "4 wrong" at rank 4 means four mismatched attributes). (c) Mismatch penalty for $q_k = 1$, where L1 is bounded by 1 and $-\log p_{gk}$ reaches 3.00 to 3.91 nats.

logits over $\mathcal{N}(g)$ via

$$\tilde{z}_g = (1 - \alpha)\, z_g + \frac{\alpha}{M} \sum_{n \in \mathcal{N}(g)} z_n, \quad (6)$$

and substitute $\tilde{z}_g$ ($\tilde{p}_g = \sigma(\tilde{z}_g)$) into Eq. (5). In Fig. 1b, single-frame ambiguity makes the raw ensemble predict Age-Old ($z_g = +0.21$), while smoothing corrects it to $\tilde{z}_g = -0.06$ ($\tilde{p}_g = 0.485$), lifting g from rank 22 to 12.

4. Experiments

We evaluate each stage of LENS under a leave-one-dataset-out protocol, test post-hoc recalibration on unseen cameras, ablate encoders, student, group CE, and kNN smoothing, and report UPAR 2027 Track 2 hidden-test results.

Evaluation protocol. We use a leave-one-dataset-out protocol across PA100k [18], PETA [8], and Market-1501 [16, 35], training on two source training splits and evaluating on the held-out validation gallery, where each distinct label vector forms a query. Held-out PETA has 6,963 images and 1,763 queries (92.6k source training images), and held-out Market-1501 has 16,458 images and 956 queries (88.7k source images). Because holding out PA100k (9,986 images, 1,479 queries) leaves only 18.7k source images (five times fewer) and yields under-trained models, we bound our full ensemble evaluation to the two high-resource folds (PETA and Market-1501) and use our archive of 48 development models across ViT and ConvNeXt encoders at sizes B, L, and H (41 on PETA, 5 on Market-1501, and 2 on

	PETA	Market
NLL on the held-out dataset (nat/image)	6.20	5.26
NLL on source datasets (nat/image)	4.48	4.28
Calibration term, held-out dataset	1.11	1.01
Calibration term, source datasets	0.80	1.07
mADM, no recalibration	69.53	74.66
mADM, oracle recalibration	71.95	76.26
mADM, source recalibration	68.51	73.65
mADM, source a_k + held-out b_k	69.41	75.54
mADM, held-out a_k + source b_k	70.39	73.77
mADM, logits divided by $T = 0.9$	69.16	74.44
mADM, logits divided by $T = 1.1$	69.64	74.79
mADM, logits divided by $T = 1.3$	69.49	74.82

Table 2. Recalibration after training for the DINOv3 + CLIP ensemble (seed 1, without WiSE-FT) using $\sigma(a_k z + b_k)$ per attribute. Oracle recalibration fits on held-out labels (in-sample diagnostic), source recalibration fits on source validation images, and rows 8–9 swap scale a_k and bias b_k. The calibration term is the NLL removed by the map fitted on the same images [12].

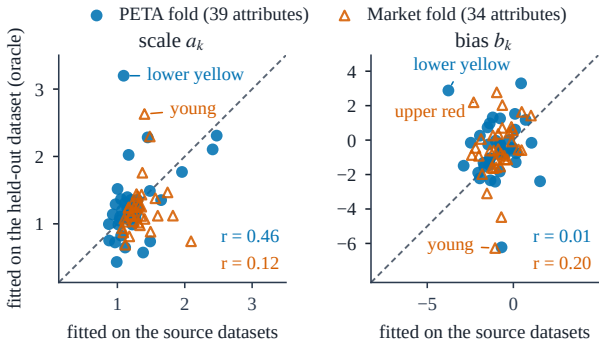


Figure 4. Per-attribute Platt parameters $\sigma(a_k z + b_k)$ fitted on source vs. held-out labels for the ensemble in Tab. 2 ($y = x$ dashed, Pearson $r = 0.46$ and 0.12 for scale, $r = 0.01$ and 0.20 for bias, omitting attributes with zero held-out positives).

PA100k) to test scoring, recalibration, and smoothing consistency across configurations rather than 48 independent benchmarks. Every query and gallery image has binary 0/1 annotations across all 40 attributes, and all 40 remain active in retrieval scoring (Eq. (5)) since the target domain is unknown at test time (zero-positive attributes, 1 on PETA and 6 on Market-1501, are masked only in BCE training and omitted in Platt fits in Fig. 4). Across 14 B-size PETA runs, attribute AUC fails to predict mADM (Spearman -0.21), whereas held-out NLL (-0.86) and listwise log-likelihood (0.89) track mADM closely and guided encoder and loss selection on PETA, while kNN hyperparameters were selected on the two source validation galleries (Tab. 5). Across seeds, replicate SD is 0.15 mADM points (bootstrap SE 0.24–0.29), and single-seed runs (Tabs. 2 and 4) match two-seed means (Tabs. 1 and 3) within 0.31 points.

Implementation details. The benchmark uses 40 binary attributes across PETA, Market-1501, and PA100k, with five mutually exclusive groups (three ages, three hair lengths, two lower-body types, 12 upper colors, and 12 lower colors). Teachers are DINOv3 ViT-H+/16 [24] (840M parameters, width 1280) and OpenAI CLIP ViT-L/14 [21] (303M, width 1024) at 224×112 resolution. We distinguish two student roles at 176×96 resolution, a compact DINOv3 ViT-B/16 (86M, width 768) for standalone CPU deployment and a DINOv3 ViT-L/16 (304M, width 1024) acting as a third ensemble member distilled with weight 0.8 using 50% mixup [34] and CutMix [33]. Each encoder freezes the patch embedding and first eight blocks, fine-tuning remaining blocks on one RTX 5090 GPU with AdamW (weight decay 0.01), batch size 64, EMA, top-block learning rate 5×10^{-5} (0.75 layer decay), and head learning rate 5×10^{-4}. CLIP WiSE-FT uses weight 0.7. Training combines DeepMAR-weighted BCE [15], label smoothing 0.05 [28], and group CE weight 0.5 for 3 teacher epochs and 10 to 20 student epochs. At inference, kNN uses 50 neighbors and weight 0.20, and probabilities are clipped to $[10^{-4}, 1 - 10^{-4}]$ before Eq. (5) so no penalty exceeds 9.21 nats. Latency is evaluated under two distinct deployment regimes rather than a cross-hardware trade-off. Offline GPU evaluation on one RTX 5090 (bfloat16) runs DINOv3 + CLIP at 2.97–3.22 ms/image plus 0.1 s for kNN and 0.8 GB RAM for float64 scoring, whereas the 6-thread CPU container (3,600 s limit for 29,248 test images) runs ViT-B/16 at 49.5 ms/image (24 min total) and ViT-L/16 at 78–103 ms.

Scoring gallery images. We first test how the scoring rule changes rankings for fixed probabilities. Table 1 compares L1 distance [31] with the likelihood score (Eq. (5)) on identical probabilities from the DINOv3 + CLIP ensemble. The likelihood score raises mADM by 7.03 points on PETA and 5.61 on Market-1501 while improving all five other metrics. Similar gains hold on standalone DINOv3 (up 5.77 and 5.48 mADM) and CLIP (up 7.33 and 5.67) teachers and across all 48 development models, beating L1 by 3.7 to 9.1 points across all 41 PETA models (mean +5.23), all 5 Market-1501 models (mean +4.94), and both PA100k models (+8.16 and +9.10).

Figure 3 shows the mechanism on a query for an adult woman. L1 ranks five frames of a man in the top five because matching 39 attributes while mispredicting female (probability 0.02–0.05) adds less than 1 to L1, smaller than mild uncertainties across 40 attributes on exact matches. The likelihood score penalizes the near-zero female probability and places four exact matches in the top five. Against related rules, literal log-DAP [14] subtracts a query-only prior constant and leaves rankings unchanged (0.00 mADM difference), whereas shifting logits by source log-priors before the sigmoid drops PETA mADM by 8.65 points, and weighting attributes by source validation AUC [25] changes

Model	PETA	Market	Mean
DINOv3	66.36	73.87	70.12
DINOv3, two fine-tuned copies	66.69	74.00	70.34
CLIP	65.81	71.72	68.76
CLIP + text score	67.17	72.32	69.74
CLIP + text score + WiSE-FT	67.38	71.98	69.68
DINOv3 + CLIP	68.92	74.46	71.69
DINOv3 + CLIP (text score)	69.49	74.79	72.14
DINOv3 + CLIP (text score, WiSE-FT)	**69.91**	**74.90**	**72.40**

Table 3. Held-out mADM (points) of individual teachers and two-teacher ensembles under the likelihood score. Row 2 averages two independently fine-tuned DINOv3 models, and the last row is the two-teacher baseline before group CE, kNN, and student.

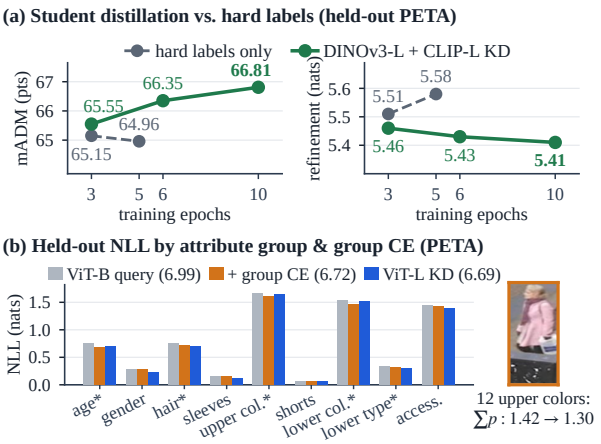


Figure 5. Improving predicted probabilities on held-out PETA. (a) Training the DINOv3 ViT-L/16 student (176 × 96) on hard labels alone vs. distilling from DINOv3-L + CLIP-L teachers across epochs. (b) Held-out NLL per image by attribute group (asterisks mark groups in Eq. (4)), with a PETA crop whose 12-color probability sum drops from 1.42 to 1.30.

mADM by at most 0.40 points across folds. Replacing independent sigmoids within each group by a group-softmax (Tab. 1) matches the likelihood score on mAP and slightly raises Rank-1 (where top-1 exact matches typically have one clear active attribute per group), yet lowers mADM by 1.22 and 0.96 points because forcing group probabilities to sum to one penalizes near matches and queries with unlabeled colors. On held-out PETA with ViT-B, adding exact expected near-match credit via Poisson-binomial DP adds only 0.29 points (62.61 to 62.90, within one SE, other domains untested).

Post-hoc recalibration on unseen cameras. The 2024 report suggested that models become over- or under-confident on new cameras [7]. We test per-attribute Platt scaling [20] (Tab. 2) in two ways. Oracle recalibration fits a per-attribute scale and bias on held-out labels as an in-sample diagnostic of target recalibration potential (since minimizing NLL in-sample is not a theoretical upper bound on mADM), whereas source recalibration fits them on source validation splits (on the seed-1 DINOv3 + CLIP ensemble without WiSE-FT, using saved source logits).

Table 2 shows that held-out miscalibration is substantial, as oracle recalibration raises mADM by 2.42 points on PETA and 1.60 on Market-1501 (and 5.73 on the two PA100k models). Yet source recalibration lowers mADM by about one point on both folds and hurts 47 models out of the 48 development models (except one linear-head PA100k model).

The failure does not occur because source cameras are well calibrated. On source validation data, the calibration term is already 72% of the held-out value in the PETA fold and exceeds it in Market-1501. While pooled 10-bin ECE on ViT-B and ViT-L rises only slightly from source validation (0.014–0.016) to held-out PETA (0.017–0.019), pooling all 40 attributes into one histogram masks opposite-sign bias shifts across attributes. Because positive and negative shifts partially cancel in the signed mean (-0.52 on PETA and -0.57 on Market-1501), the macro-averaged absolute bias is more than twice as large at 1.24 and 1.34. Rather, per-attribute shift directions differ across cameras (Fig. 4). Platt scale stays above 1 for most attributes on both splits and changes little across cameras (median change below 0.18, rank correlation 0.41 and 0.34), whereas per-attribute bias is nearly uncorrelated between source and held-out cameras (95% CIs spanning zero, median change 0.92 and 0.81, and roughly one third flipping sign).

Swapping scale and bias between domains (Tab. 2, rows 8–9) confirms that this bias shift drives the degradation. Across all 48 development runs, pairing source scale with held-out bias recovers 61%–90% of the oracle NLL reduction across the three folds (vs. 22%–33% for held-out scale with source bias), raises Market-1501 mADM by 0.88 points over the uncalibrated model, and removes nearly all source-calibration loss on PETA (though reaching the full PETA oracle requires both held-out parameters). Conversely, keeping source bias with oracle scale still degrades Market-1501 by 0.89 points below the uncalibrated model. Because per-attribute biases are potentially associated with label prevalence, rare positives, and annotation conventions across cameras, applying source biases shifts gallery images in the wrong direction and reorders rankings even when average NLL drops (6.20 to 5.94 nats on PETA). Dividing logits by a shared temperature (Tab. 2) shifts mADM by at most 0.37 points, so LENS uses raw logits without post-hoc recalibration.

The role of two different encoders. Because probabilities cannot be fixed after training, we improve the learned representations (Tab. 3). On held-out PETA under Eq. (5), the UPAR ConvNeXt-B baseline [6, 7] (Fig. 6a) reaches 59.59 mADM (62.26 for ConvNeXt-L), and single-encoder query-head models with AIMv2-L, SigLIP2-L, or

DINOv3 + CLIP ensemble	PETA	Market	Mean
Likelihood score	69.84	74.59	72.22
+ kNN smoothing	70.23	75.31	72.77
+ group CE	70.89	75.28	73.08
+ group CE + kNN smoothing (submitted)	**71.03**	75.66	73.35
kNN with pretrained DINOv3	71.00	75.52	73.26
+ query expansion	70.99	**75.94**	**73.47**

Table 4. Teacher-only factorial ablation (held-out mADM in points) of DINOv3 + CLIP (seed 1) under Eq. (5), whereas the three-model ensemble with the student is evaluated on PETA and the hidden test set in Sec. 4 and Tab. 6. Group CE applies to both teachers, kNN uses $M = 50$, $\alpha = 0.2$, and fine-tuned DINOv3 features except row 5, and row 1 uses CLIP seed 1 and a retrained DINOv3.

Neighbor feature	M, α	PETA	Market	PA100k
no smoothing (mADM)		68.45	78.14	73.86
DINOv3, mean patch token	3, 0.25	−0.03	+0.67	+0.52
DINOv3, mean patch token	10, 0.25	+0.56	+0.86	+0.67
DINOv3, mean patch token	20, 0.25	+0.64	+0.86	+0.65
DINOv3, mean patch token	50, 0.25	+0.60	+0.85	+0.70
DINOv3, mean patch token	50, 0.2	+0.70	+0.69	+0.66
DINOv3, mean patch token	20, 0.5	−1.58	+0.82	−0.29
DINOv3, CLS token	20, 0.25	+0.54	+0.64	+0.78
CLIP, mean patch token	20, 0.25	+0.23	+1.07	+0.29

Table 5. Change in mADM (points) from Eq. (6) relative to row 1 on an earlier DINOv3 + CLIP ensemble of the PETA fold (without text score) across held-out PETA and the two source validation galleries (Market-1501 and PA100k, excluded from training).

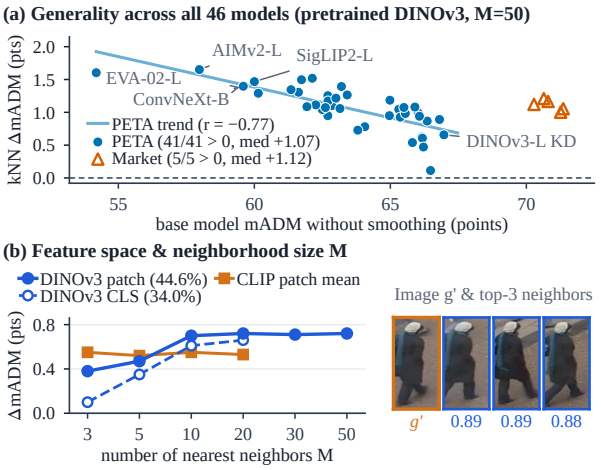


Figure 6. Generality and feature analysis of kNN smoothing. (a) ΔmADM across all 46 development models with saved gallery features ($M = 50$, $\alpha = 0.2$) using pretrained DINOv3 features, where all 41 PETA models ($[+0.11, +1.65]$, median $+1.07$) and all 5 Market-1501 models ($[+1.00, +1.21]$, median $+1.12$) improve. (b) Mean ΔmADM across Tab. 5 galleries vs. M ($\alpha = 0.25$), with g' and top-3 neighbors.

MetaCLIP2-H reach 57.98–63.15, all below DINOv3 at 66.36. In Tab. 3, averaging two fine-tuned copies of DINOv3 adds only 0.22 mean mADM over one copy, whereas pairing DINOv3 with CLIP adds 1.57 points even though CLIP alone is weaker than DINOv3. While WiSE-FT slightly lowers standalone CLIP on Market-1501, combining the CLIP text score and WiSE-FT with DINOv3 improves both folds (up 0.71 mean mADM over vanilla DINOv3 + CLIP) by tempering overconfident CLIP logits. Training both encoders jointly with one shared query head and group CE reaches 69.92 on PETA and 75.04 on Market-1501, trailing the independent two-teacher ensemble in Tab. 4 by 0.97 and 0.24 points.

To isolate distillation across training length without mixup or CutMix (Fig. 5a), we train the DINOv3 ViT-L/16 student on the PETA fold from DINOv3-L and CLIP-L teachers (as the fold student was trained only on PETA before full-data training). At 3 epochs, distillation improves mADM by 0.40 points over hard labels, and extending hard-label training to 5 epochs overfits and worsens refinement, whereas distillation steadily improves through 10 epochs to 66.81 mADM (and 67.51 with 20-epoch mixup and CutMix, or up 1.71 points on ViT-B/16). Adding this 10-epoch ViT-L/16 student to DINOv3 + CLIP raises held-out PETA mADM from 69.91 to 70.24 (and on the earlier PETA ensemble from 68.45 to 69.21, rising to 69.73 with kNN).

Group CE and kNN smoothing. Because the likelihood score depends on probability quality, we regularize mutually exclusive groups in training with group CE (Fig. 5b). Over 92% of held-out queries have a single active positive in four of the five groups (77.1% on PETA upper color, 81.5% on Market-1501 hair length), while in training Eq. (4) skips 11.7% of PETA upper-color and 12.2% of PA100k hair labels without a unique positive. Across eight seed-matched single-model pairs differing only in group CE (six on PETA, two on Market-1501), group CE raises mADM in all eight by 0.40 to 1.25 points (and adds 1.30 and 1.03 points on standalone DINOv3 ViT-H+/16). In Fig. 5b, the five regularized groups account for 94% of the held-out PETA NLL reduction, split evenly between the two color groups and the age, hair, and lower-body groups, aligning predictions with single-label conventions and curbing conflicting probabilities within a group (as on the crop whose 12-color probability sum drops from 1.42 to 1.30). On PETA, nearly all of the NLL drop (0.17 to 0.27 nats) comes from the calibration term (0.15 to 0.23 nats), whereas on Market-1501 it mainly improves refinement.

On a single DINOv3 ViT-B/16 model, group CE and kNN smoothing stack cleanly, adding 1.20 and then 0.78 mADM points on PETA (0.56 and then 1.05 on Market-1501). On the two-teacher ensemble (Tab. 4), group CE adds 1.05 points on PETA and 0.69 on Market-1501, while kNN smoothing adds 0.39 and 0.72 alone, and 0.14 (within

Submission	mADM
DINOv3 ViT-B/16 student	63.65
DINOv3 + CLIP + DINOv3 ViT-L/16 student	67.22
+ text score, WiSE-FT, kNN smoothing	67.83
+ group CE for DINOv3 and CLIP	**68.17**

Table 6. Sequential submissions (mADM in points) on the UPAR 2027 Track 2 hidden test set under Eq. (5) (row 3 reuses row 2 DINOv3 and student weights while adding CLIP text score, WiSE-FT, and kNN, Tabs. 3 and 4, with 39.52 mAP, 27.91 mINP, 40.79 R-1, 61.45 R-5, 69.91 R-10, and row 4 adds group CE).

single-seed noise of 0.15) and 0.38 after group CE. Frozen pretrained DINOv3 features give nearly the same kNN gain on PETA and two thirds on Market-1501 without fine-tuning, whereas query expansion [5] helps Market-1501 mADM but lowers Rank-5 and Rank-10.

Sensitivity of kNN smoothing. Table 5 evaluates kNN smoothing across neighbor count, weight, and feature type on an earlier DINOv3 + CLIP ensemble across held-out PETA and the two source validation galleries (Market-1501 and PA100k, excluded from training). Selecting neighbor count and weight on the two source validation galleries alone identifies a conservative plateau across 10 to 50 neighbors and weights of 0.20 to 0.25 that transfers to held-out PETA (up 0.56 to 0.70 mADM in Tab. 5) and held-out Market-1501 (0.73 to 0.83).

By contrast, an aggressive weight of 0.50 reveals how gallery structure governs negative transfer (Tab. 5). Under DINOv3 features, the closest neighbor shares all 40 ground-truth labels with the image for 76.7% of Market-1501, 50.6% of PA100k, and 44.6% of PETA images (Fig. 1b, Fig. 6b), as 85% of Market-1501 queries have multi-frame tracklets whereas 61% of PETA queries have one gallery match. Thus, on multi-shot Market-1501, 3 neighbors help (up 0.67) and weight 0.50 stays within same-identity tracklets (up 0.82), whereas on sparse-identity PETA (and PA100k), 3 neighbors do not help (down 0.03) and weight 0.50 causes negative transfer (down 1.58 and 0.29) by over-weighting distractors that differ on fine attributes.

Three controls confirm that conservative smoothing requires visual neighborhoods rather than generic logit shrinkage. Uniformly softening logits with temperature 1.1 or 1.3 (Tab. 2) changes PETA mADM by at most 0.11 points and hurts PA100k by up to 1.34, finding neighbors by predicted probability cosine lowers PETA mADM by 0.13 points (up 0.17 on Market-1501), and restricting smoothing to k-reciprocal mutual neighbors [36] helps multi-shot Market-1501 (up 0.86) but hurts sparse-identity PETA (down 0.32) by starving single-match frames, whereas DINOv3 kNN improves both (up 0.44 and 0.73). With frozen pretrained DINOv3 features, kNN smoothing needs no training (Fig. 6a) and improves all 46 development models with saved gallery features across all four encoder families (DINOv3 ViT-B and ViT-L, ConvNeXt, and CLIP/SigLIP2/AIMv2/EVA-02), raising all 41 PETA models (mean +1.06, 95% CI 0.96 to 1.15) and all 5 Market-1501 models (mean +1.11, CI 1.05 to 1.17), with larger gains on weaker models (correlation -0.77).

Evaluation on the UPAR 2027 hidden test set. For hidden-test submissions (Tab. 6, where no 2027 baseline was released), teachers and students are trained on all training and validation splits of PA100k, PETA, and Market-1501 with the fold hyperparameters, separate from held-out ablations. Within the CPU budget, the standalone 86M DINOv3 ViT-B/16 with group CE reaches 62.47 mADM, rising to 63.15 with 10-epoch distillation and 63.65 with 20-epoch mixup and CutMix. Combining DINOv3 ViT-H+/16, CLIP ViT-L/14, and the 304M DINOv3 ViT-L/16 student (Fig. 1) reaches 67.22 mADM. Reusing those DINOv3 and student weights while adding CLIP text score, WiSE-FT, and kNN reaches 67.83 (Tabs. 3 and 4), and retraining teachers with group CE (+0.34) yields **68.17**.

5. Limitations and future work

First, Eq. (5) assumes conditional independence and ranks by exact-match probability rather than expected mADM (tested via Poisson-binomial DP only on PETA with ViT-B), while our three-model ensemble and factorial ablations are bounded to the two high-resource folds rather than the source-scarce PA100k fold (and repeated PETA evaluation risks adaptive selection bias, alongside 20-epoch student runs that bundle mixup and CutMix). Second, oracle Platt scaling is an in-sample diagnostic of target recalibration potential rather than a theoretical mADM bound, and we do not regress bias shifts on prevalence shifts, split target galleries for calibration, or ablate demographic vs. color group CE. Third, kNN smoothing is transductive and sensitive to gallery redundancy, so random-neighbor, one-frame-per-identity, and same-identity-exclusion controls plus similarity weights are needed to isolate cross-identity smoothing.

6. Conclusion

We studied attribute-based person retrieval on unseen cameras across PA100k, PETA, Market-1501, and UPAR 2027. Scoring gallery images by exact-query log-likelihood beats L1 distance on all six metrics for our ensemble and across all 48 tested model configurations by penalizing confident mismatches. Parameter swaps show that while target recalibration raises mADM and Platt scale transfers across cameras, per-attribute biases shift in uncorrelated directions so source-fitted Platt maps lower mADM even as average NLL drops. Across the high-resource folds and hidden test set, combining DINOv3, CLIP, student distillation, group CE, and kNN reaches 68.17 mADM.

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